

Module 1:

Problem Statement:

Want to feel better, have more energy and even add years to your life? Just exercise. Joining a gym can help you stay motivated to exercise consistently. This is a great way to build muscle, lose weight, lower blood pressure, boost mental focus, and more. Overtime, you can look better, feel better, and accomplish things you never thought possible! It is important to know your BMI (Body Mass Index) for the good health. (for more read below)

Write a C/C++ program to process the Gym data using the following constraints:

- i. Store ID, Height and Weight of each member
- ii. A member can be added/removed/updated
- iii. The program should be menu operated
- iv. Define a structure with data members ID, Height and Weight.
- v. Calculate average Height of the members
- vi. Calculate average Weight of the members
- vii. Calculate Max Height and Weight
- viii. Calculate Min Height and Weight
- ix. Display BMI classification of a given member (use following table)

Code:

```
#include<bits/stdc++.h>
#define ll long long int
#define yess cout<<"YES\n"
#define noo cout<<"NO\n"
#define endl "\n"
#define newl cout<<endl
#define clr system("cls")
using namespace std;
const int N=1e4;
int main();
void classify(float x)
{
    if(x>40){ cout<<"Obese Class III"; }
    else if(x>=35){ cout<<"Obese Class II"; }
    else if(x>=30){ cout<<"Obese Class I"; }
    else if(x>=25){ cout<<"Overweight"; }
    else if(x>=18.5){ cout<<"Normal"; }
    else if(x>=17){ cout<<"Mild Thinness"; }
    else if(x>=16){ cout<<"Moderate Thinness"; }
    else{ cout<<"Severe Thinness"; }
}
void menu()
{
    cout<<"***** Main Menu *****\n";
    cout<<"1. Add member\n2. Update member\n3. Remove member\n";
    cout<<"4. Max height & weight\n5. Min height & weight\n";
    cout<<"6. Average height & weight\n7. BMI classification\n";
    cout<<"8. Show member's Height and weight\n";
    cout<<"9. Found Member\n10. Exit\n";
    cout<<"Enter your option(1-10):_ ";
}
```

//Terminal:

```
***** Main Menu *****
1. Add member
2. Update member
3. Remove member
4. Max height & weight
5. Min height & weight
6. Average height & weight
7. BMI classification
8. Show member's Height and weight
9. Found Member
10. Exit
Enter your option(1-10):_
```

```
bool check_height(float x)
{
    if(x<0.30 | | x>2.5)
    {
        cout<<"Height should be between 0.3 to 2.5 meter\n";
        cout<<"\nPress 1 to back\n";
        abar:
        int b;cin>>b;
        if(b==1)
        {
            newl; return true;
        }
        else
        {
            clr;
            cout<<"Press correct keyword 1\n"; goto abar;
        }
    }
    return false;
}

bool check_weight(float x)
{
    if(x<3 | | x>500)
    {
        cout<<"weight should be between 3 to 500 KG\n";
        cout<<"\nPress 1 to back\n";
        abar1:
        int b;cin>>b;
        if(b==1)
        {
            newl; return true;
        }
        else
        {
            clr;
            cout<<"Press correct keyword 1\n"; goto abar1;
        }
    }
    return false;
}

struct member
{
```

```

int member_id;
float Height;
float Weight;
};
struct member Gym[N],Gym1 [N],Gym2[N],Gym3[N],Gym4[N];
void Add()
{
    begin:
    cout<<"Enter your member id: ";
    int a;cin>>a;
    if(Gym[int(a)].member_id==0)
    {
        hoyni:
        cout<<"Enter your height (Meter): ";
        float x;cin>>x;
        if(check_height(x))
        {
            clr; menu(); goto hoyni;
        }
        hoyni1:
        cout<<"Enter your weight (KG): ";
        float y;cin>>y;
        if(check_weight(y))
        {
            clr; menu(); goto hoyni1;
        }
        Gym[a].member_id=a; Gym[a].Height=x; Gym[a].Weight=y;
        Gym1[a].member_id=a; Gym1[a].Height=x; Gym1[a].Weight=y;
        Gym2[a].member_id=a; Gym2[a].Height=x; Gym2[a].Weight=y;
        Gym3[a].member_id=a; Gym3[a].Height=x; Gym3[a].Weight=y;
        Gym4[a].member_id=a; Gym4[a].Height++; Gym4[a].Weight++;
        cout<<"\nmember added successfully\n\nPress 8 to clear\n";
        flag:
        int b;cin>>b;
        if(b==8)
        {
            clr; main();
        }
        else
        {
            clr; menu(); newl;
            cout<<"Enter correct keyword 8\n"; goto flag;
        }
    }
    else
    {
        cout<<"Member id is already registered or Invalid Key word\nPress 8 to clear\n";
        again:
        int b;cin>>b;
        if(b==8)
        {
            clr; menu(); newl; goto begin;
        }
        else
        {

```

```

        clr; menu(); newl;
        cout<<"Enter correct keyword 8\n"; goto again;
    }
}
}
//Terminal:

```

```

Enter your option(1-9):_ 1
Enter your member id: 1
Enter your height (Meter): 1.55
Enter your weight (KG): 57

member added successfully

Press 8 to clear

```

```

void update()
{
    begin1:
    cout<<"Enter your member id: ";
    int a;cin>>a;
    if(Gym[int(a)].member_id!=0)
    {
        cout<<"Which you want to update\nPress 0 for height\nPress 1 for weight\nPress 2 for both\nyour
option->";
        int m;cin>>m;
        if(m==0)
        {
            hoyni2:
            cout<<"Present height->"<<Gym[a].Height; newl;
            cout<<"Enter your updated height (Meter): "; float x;cin>>x;
            if(check_height(x))
            {
                clr; menu(); goto hoyni2;
            }
            Gym[a].Height=x; Gym1[a].Height=max(Gym1[a].Height,x);
            Gym2[a].Height=min(Gym2[a].Height,x); Gym3[a].Height+=x; Gym4[a].Height++;
            cout<<"\nHeight updated successfully\n\nPress 8 to clear\n";
        }
        else if(m==1)
        {
            hoyni3:
            cout<<"Present weight->"<<Gym[a].Weight;
            cout<<"\nEnter your updated weight (KG): ";
            float y;cin>>y;
            if(check_weight(y))
            {
                clr; menu(); goto hoyni3;
            }
            Gym[a].Weight=y; Gym1[a].Weight=max(Gym1[a].Weight,y);
            Gym2[a].Weight=min(Gym2[a].Weight,y); Gym3[a].Weight+=y;
            Gym4[a].Weight++;
            cout<<"\nWeight updated successfully\n\nPress 8 to clear\n";
        }
        else if(m==2)
        {

```

```

hoyni4:
cout<<"Present height->"<<Gym[a].Height;newl;
cout<<"Enter your updated height (Meter): ";
float x;cin>>x;
if(check_height(x))
{
    clr; menu(); goto hoyni4;
}
Gym[a].Height=x;Gym1[a].Height=max(Gym1[a].Height,x);
Gym2[a].Height=min(Gym2[a].Height,x); Gym3[a].Height+=x;
Gym4[a].Height++;
hoyni5:
cout<<"Present weight->"<<Gym[a].Weight;
cout<<"\nEnter your updated weight (KG): ";
float y;cin>>y;
if(check_weight(y))
{
    clr; menu(); goto hoyni5;
}
Gym[a].Weight=y; Gym1[a].Weight=max(Gym1[a].Weight,y);
Gym2[a].Weight=min(Gym2[a].Weight,y); Gym3[a].Weight+=y;
Gym4[a].Weight++;
cout<<"Height and Weight updated successfully\n\nPress 8 to clear\n";
}
flag1:
int b;cin>>b;
if(b==8)
{
    clr; main();
}
else
{
    clr; menu();newl;
    cout<<"Enter correct keyword 8\n"; goto flag1;
}
}
else
{
    cout<<"member id is not found or Invalid Key word\nPress 8 to back\n";
    again1:
    int b;cin>>b;
    if(b==8)
    {
        clr; menu(); newl; goto begin1;
    }
    else
    {
        clr; menu(); newl;
        cout<<"Enter correct keyword 8\n"; goto again1;
    }
}
}
}
//Terminal:

```

```

Enter your option(1-9):_ 2
Enter your member id: 1
Which you want to update
Press 0 for height
Press 1 for weight
Press 2 for both
your option->2
Present height->1.55
Enter your updated height (Meter): 1.56
Present weight->57
Enter your updated weight (KG): 58
Height and Weight updated successfully
Press 8 to clear

```

```

void remove()
{
    begin2:
    cout<<"Enter your member id: ";
    int a;cin>>a;
    if(Gym[int(a)].member_id!=0)
    {
        Gym[a]={0,0,0};Gym1[a]={0,0,0};Gym2[a]={0,0,0};Gym3[a]={0,0,0};Gym4[a]={0,0,0};
        cout<<"\nremoved successfully\n\nPress 8 to clear\n";
        flag2:
        int b;cin>>b;
        if(b==8)
        {
            clr; main();
        }
        else
        {
            clr; menu(); newl;
            cout<<"Enter correct keyword 8\n"; goto flag2;
        }
    }
    else
    {
        cout<<"member id is not found or Invalid Key word\nPress 8 to back\n";
        again2:
        int b;cin>>b;
        if(b==8)
        {
            clr; menu(); newl; goto begin2;
        }
        else
        {
            clr; menu(); newl;
            cout<<"Enter correct keyword 8\n"; goto again2;
        }
    }
}

```

//Terminal:

```

Enter your option(1-10):_ 3
Enter your member id: 1

removed successfully

Press 8 to clear

```

```

void Mh_Mw()
{
    begin3:
    cout<<"Enter your member id: ";
    int a;cin>>a;
    if(Gym[int(a)].member_id!=0)
    {
        cout<<"Maximum height->"<<Gym1[a].Height<<endl;
        cout<<"Maximum weight->"<<Gym1[a].Weight;
        newl; cout<<"\n\nPress 8 to clear\n";
        flag3:
        int b;cin>>b;
        if(b==8)
        {
            clr; main();
        }
        else
        {
            clr; menu(); newl;
            cout<<"Enter correct keyword 8\n"; goto flag3;
        }
    }
    else
    {
        cout<<"member id is not found or Invalid Key word\nPress 8 to back\n";
        again3:
        int b;cin>>b;
        if(b==8)
        {
            clr; menu(); newl; goto begin3;
        }
        else
        {
            clr; menu(); newl;
            cout<<"Enter correct keyword 8\n"; goto again3;
        }
    }
}

```

//Terminal:

```

Enter your option(1-9):_ 4
Enter your member id: 1
Maximum height->1.56
Maximum weight->58

Press 8 to clear

```

```

void mh_mw()
{
    begin4:
    cout<<"Enter your member id: ";
    int a;cin>>a;
    if(Gym[int(a)].member_id!=0)
    {
        cout<<"Minimum height->"<<Gym2[a].Height<<endl;
        cout<<"Minimum weight->"<<Gym2[a].Weight;
    }
}

```

```

newl; cout<<"\n\nPress 8 to clear\n";
flag4:
int b;cin>>b;
if(b==8)
{
    clr; main();
}
else
{
    clr; menu(); newl;
    cout<<"Enter correct keyword 8\n"; goto flag4;
}
}
else
{
    cout<<"member id is not found or Invalid Key word\nPress 8 to back\n";
again4:
int b;cin>>b;
if(b==8)
{
    clr; menu(); newl; goto begin4;
}
else
{
    clr; menu(); newl; cout<<"Enter correct keyword 8\n";
    goto again4;
}
}
}
}
//Terminal:

```

```

Enter your option(1-9):_ 5
Enter your member id: 1
Minimum height->1.55
Minimum weight->57

Press 8 to clear

```

```

void Ah_Aw()
{
    begin5:
    cout<<"Enter your member id: ";
    int a;cin>>a;
    if(Gym[int(a)].member_id!=0)
    {
        cout<<"Average height->"<<(Gym3[a].Height/Gym4[a].Height)<<endl;
        cout<<"Average weight->"<<(Gym3[a].Weight/Gym4[a].Weight);
        newl; cout<<"\n\nPress 8 to clear\n";
        flag5:
        int b;cin>>b;
        if(b==8)
        {
            clr; main();
        }
        else
        {

```



```

        clr; menu(); newl;
        cout<<"Enter correct keyword 8\n"; goto flag5;
    }
}
else
{
    cout<<"member id is not found or Invalid Key word\nPress 8 to back\n";
    again5:
    int b;cin>>b;
    if(b==8)
    {
        clr; menu(); newl; goto begin5;
    }
    else
    {
        clr; menu(); newl;
        cout<<"Enter correct keyword 8\n"; goto again5;
    }
}
}
}

```

//Terminal:

```

Enter your option(1-9):_ 6
Enter your member id: 1
Average height->1.555
Average weight->57.5

```

```

void BMI()
{
    begin6:
    cout<<"Enter your member id: ";
    int a;cin>>a;
    if(Gym[a].member_id!=0)
    {
        cout<<"Your BMI is->";
        float f=Gym[a].Weight/(Gym[a].Height*Gym[a].Height);
        cout<<f; newl;
        cout<<"Your BMI classification-->"; classify(f);
        cout<<"\n\nPress 8 to clear\n";
        flag6:
        int b;cin>>b;
        if(b==8)
        {
            clr; main();
        }
        else
        {
            clr; menu(); newl;
            cout<<"Enter correct keyword 8\n"; goto flag6;
        }
    }
}
else
{
    cout<<"member id is not found or Invalid Key word\nPress 8 to back\n";
    again6:
    int b;cin>>b;
}
}

```

```

        if(b==8)
        {
            clr; menu(); newl; goto begin6;
        }
        else
        {
            clr; menu(); newl;
            cout<<"Enter correct keyword 8\n";goto again6;
        }
    }
}

```

//Terminal:

```

Enter your option(1-9):_ 7
Enter your member id: 1
Your BMI is->23.833
Your BMI classification--->Normal

Press 8 to clear

```

```

void show()
{
    begin8:
    cout<<"Enter your member id: ";
    int a;cin>>a;
    if(Gym[a].member_id!=0)
    {
        cout<<"\nYour member id->"<<a;
        cout<<"\nYour Height is->"<<Gym[a].Height;
        cout<<"\nYour Weight is->"<<Gym[a].Weight;
        cout<<"\n\nPress 8 to clear\n";
        flag8:
        int b;cin>>b;
        if(b==8)
        {
            clr; main();
        }
        else
        {
            clr; menu(); newl;
            cout<<"Enter correct keyword 8\n"; goto flag8;
        }
    }
    else
    {
        cout<<"member id is not found or Invalid Key word\nPress 8 to back\n";
        again8:
        int b;cin>>b;
        if(b==8)
        {
            clr; menu(); newl; goto begin8;
        }
        else
        {
            clr; menu(); newl;
            cout<<"Enter correct keyword 8\n"; goto again8;
        }
    }
}

```

```
}  
}
```

//Terminal:

```
Enter your option(1-10):_ 8  
Enter your member id: 1
```

```
Your member id->1  
Your Height is->1.56  
Your Weight is->58  
  
Press 8 to clear
```

```
void exit()  
{  
    clr; cout<<"Press 1 to restart the enquiry again-->";  
    flag7:  
    int b;cin>>b;  
    if(b==1)  
    {  
        clr; main();  
    }  
    else  
    {  
        clr; cout<<"Enter correct keyword 9-->"; goto flag7;  
    }  
}
```

//Terminal:

```
Press 1 to restart the enquiry again-->|
```

```
void found()  
{  
    begin8:  
    cout<<"Enter your member id: ";  
    int a;cin>>a;  
    if(Gym[a].member_id!=0)  
    {  
        cout<<"\nMember Found\n"; cout<<"\n\nPress 8 to clear\n";  
        flag8:  
        int b;cin>>b;  
        if(b==8)  
        {  
            clr; main();  
        }  
        else  
        {  
            clr; menu(); newl;  
            cout<<"Enter correct keyword 8\n"; goto flag8;  
        }  
    }  
    else  
    {  
        cout<<"member is Not Found\nPress 8 to back\n";  
        again8:  
        int b;cin>>b;
```

```

        if(b==8)
        {
            clr; menu(); newl; goto begin8;
        }
        else
        {
            clr; menu(); newl;
            cout<<"Enter correct keyword 8\n"; goto again8;
        }
    }
}
//terminal:

```

```

Enter your option(1-10):_ 9
Enter your member id: 1

Member Found

Press 8 to clear

```

```

void rej()
{
    clr;
    cout<<"Invalid keyword\n";
    cout<<"Press 9 to continue\n";
    flag9:
    int b;
    cin>>b;
    if(b==9)
    {
        clr; main();
    }
    else
    {
        clr; cout<<"Enter correct keyword 9-->";goto flag9;
    }
}

int main()
{
    menu();
    while(1)
    {
        int i;cin>>i;
        if(i==1){ Add(); }
        else if(i==2){ update(); }
        else if(i==3){ remove(); }
        else if(i==4){ Mh_Mw(); }
        else if(i==5){ mh_mw(); }
        else if(i==6){ Ah_Aw(); }
        else if(i==7){ BMI(); }
        else if(i==8){ show(); }
        else if(i==9){ found(); }
        else if(i==10){ exit(); }
        else{ rej(); }
    }
}

```